

Predicting Airline Passenger Satisfaction: A Machine Learning Approach to Enhance Customer Retention

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1 Business Scope

1.1 Problem Definition

The airline industry is a highly competitive environment characterized by thin profit margin and high price sensitivity. In such a context, customer retention is a critical strategic lever. Industry studies estimate that acquiring a new customer is 5 to 25 times more expensive than retaining an existing one.

Passengers satisfaction is the primary driver of loyalty. A dissatisfied passenger is not only more likely to switch to a competitor but can also damage the brand's reputation through negative reviews and social media. The core problem this project addresses is the reactive nature of traditional customer service: airline often realize a customer is unhappy only after they have filed a complaint or even stopped booking.

1.2 Objective

Our project aims to shift from a negative to a proactive approach by developing and applying a predictive Machine Learning model. By analyzing flight data and post-flight surveys, we aim to achieve three main goals:

- **Proactive Retention** : Accurately classify passengers into "Satisfied" or "Neutral/Dissatisfied" categories to identify at risk customers before they churn. This allows the airline to target them with personalized compensation or care.
- **Service Quality Improvement** : Identify the key drivers of dissatisfaction. For instance, determining if "In-flight WiFi" or "Seat Comfort" has a higher impact on the Score, and help prioritize investment in cabin upgrades.
- **Operational Efficiency** : Quantify the impact of operational metrics, such as Departure Delay and Arrival Delay, on perceived satisfactions. This supports data-driven decision making regarding logistics and ground operation.

1.3 Alignment with Major (Data Science and AI)

While the business goals focus on retention, the implementation of this project strictly aligns with our major in **Data Science and Artificial Intelligence**.

Unlike standard statistical analysis, our approach focuses on engineering a robust, end-to-end Machine Learning pipeline capable of deployment. Our technical contributions include:

1.3.1 Benchmarking and Complexity

We move beyond simple baselines to rigorously compare linear models (Logistic Regression) against non-linear ensemble methods (Random Forest) and state-of-art Gradient Boosting techniques (XGBoost). This comparison highlights the non-linear decision boundaries inherent in human satisfaction data.

1.3.2 Advanced Data Handling

We address real-world challenges, specifically :

- Handling mixed data types (numerical data and categorical survey data) using encoding pipelines.
- Managing potential class imbalance to ensure the model does not simply favor the majority class, optimizing for **F1-Score** and **AUC-ROC** rather than just **Accuracy**.

1.3.3 State-of-the-Art Implementation

By integration **XGBoost** as an "Out-of-scope" algorithm, we demonstrate our ability to leverage advanced regularization and gradient-based optimization to maximize predictive performance.

2 Data Description and Exploration

2.1 Dataset Overview

The dataset utilized in this project is the *Airline Passenger Satisfaction* dataset (source: Kaggle). It consists of customer feedback and flight details from a major airline.

- **Dimensions** : The training set contains 103,904 observations and 24 features.
- **Target Variable** : The `satisfaction` column acts as our binary target, classifying passengers as either "satisfied" or "neutral or dissatisfied".

2.2 Feature Dictionary

The features can be categorized into three main groups:

2.2.1 Customer Demographics:

- *Gender* : Male or Female.
- *Age* : The age of the passenger.
- *Customer Type* : Differentiates between "Loyal" and "Disloyal" customers.

2.2.2 Flight Characteristics:

- *Type of Travel* : Purpose of the trip (Business vs. Personal).
- *Class* : Travel class (Business, Eco, Eco Plus).
- *Flight Distance* : The distance of the flight in miles.

2.2.3 Service Ratings and Operations:

- **Satisfaction Drivers** : 14 distinct services (*In-flight WiFi service*, *Seat comfort*, *Leg room service*) rated on a scale from 0 to 5.
- **Operational Metrics** : *Departure Delay in Minutes* and *Arrival Delay in Minutes*.

2.3 Exploratory Data Analysis (EDA)

Before modeling, we performed a thorough analysis to understand data quality and distributions.

- **Target Distribution** : We visualized the balance of the target using a pie chart. The data set is relatively balanced, with approximately 43% of the passengers being "satisfied" and 57% being "neutral or dissatisfied". This slight imbalance does not require aggressive resampling technique like SMOTE, but warrants the use of metrics like F1-Score alongside Accuracy.

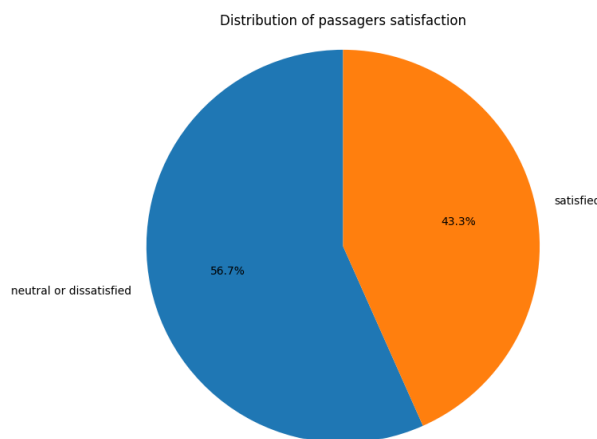


Figure 1: Pie Chart of data balance

- **Missing Values Analysis** : We checked for null values across all columns. The dataset is mostly clean, with the exception of the *Arrival Delay in Minutes* feature, which contained 310 missing values (approx 0.3% of the data).

- **Interpretation** : These missing values likely correspond to flights with no delay data recorded or canceled flights.
- **Strategy** : Given the small proportion of missing data, we opted for **Median Imputation** in our preprocessing pipeline to preserve the data distribution without discarding valuable rows

```

--- Missing values count ---
Gender 0
Customer Type 0
Age 0
Type of Travel 0
Class 1
Flight Distance 1
Inflight wifi service 1
Departure/Arrival time convenient 1
Ease of Online booking 1
Gate location 1
Food and drink 1
Online boarding 1
Seat comfort 1
Inflight entertainment 1
On-board service 1
Leg room service 1
Baggage handling 1
Checkin service 1
Inflight service 1
Cleanliness 1
Departure Delay in Minutes 1
Arrival Delay in Minutes 297
satisfaction 1
dtype: int64

```

Figure 2: Missing values

- **Correlation Insights** : A preliminary correlation analysis revealed strong multicollinearity between *Departure Delay* and *Arrival Delay* (correlation near 0.96). However, we decided to keep both features as they might carry distinct operational signals for the model.

3 Methodology (Pre-processing)

To ensure a robust, reproducible, and deployment-ready workflow, we implemented a structured preprocessing strategy using **Scikit-Learn Pipelines**. This approach is critical to prevent data leakage (where information from the test set leaks into the training process) and simplifies the transformation of new data.

3.1 Data Cleaning and Preprocessing

Our preprocessing pipeline distinguishes between numerical and categorical features, applying specific transformations to each types via a *Column Transformer*:

3.1.1 Numerical feature Pipeline

- **Imputation** : We applied **Median Imputation** to handle missing values (specifically for the **Arrival Delay in Minutes** feature as we saw). We chose the median over the mean because it is more robust to outliers, which are common in flight delay data.

- **Scaling** : We used **Standard Scaler** to normalize the data (mean=0, variance = 1). This step is mandatory for distance based algorithms and linear model (Logistic Regression) to ensure that features with a large magnitudes (*Flight Distance*) do not dominate the objective function during gradient descent.

3.1.2 Categorical Features Pipeline

- **Imputation** : Missing categorical values filled using the **Most Frequent** value.
- **Encoding** : We utilized **One Hot Encoder** to transform categorical variables (*Gender, Class, Type of Travel*) into binary vectors. We set the following parameter `handle_unknown='ignore'` to ensure the model remains robust in production if it encounters unseen categories.

3.2 Data Splitting Strategy

Before any preprocessing took place, we split the dataset to simulate a real-world scenario and strictly evaluate generalization performance.

- **Train/Test Ratio** : We allocated **80%** of the data for the training and **20%** for testing.
- **Reproducibility** : We set a fixed `random_state = 42` to ensure that our results are consistent and reproducible across different runs.

3.3 Feature Engineering (PCA)

To address the high multicollinearity observed during exploration (specifically between *Departure Delay* and *Arrival Delay*, we integrated **Principal Analysis Component (PCA)** step into our pipeline.

- **Implementation** : We applied PCA to the scaled numerical features, dynamically selecting the number of components required to preserve **95% of the explained variance**.
- **Objective** : The goal was to test if reducing the feature space could improve model training speed and generalization without significantly sacrificing predictive accuracy.

3.4 Validation Protocol: Cross-Validation

A simple train/test split can sometimes lead to biased results depending on how the data is randomly divided. To ensure the statistical robustness of our findings, we adopted a **K-Fold Cross-Validation** strategy.

- **Configuration** : We used a **5-Fold Cross-Validation** on our champion models. This involves splitting the data into 5 subsets, training on 4 and validation on the 5th, rotating this process 5 times.
- **benefit** : This provides a mean accuracy score and a standard deviation, allowing us to verify that our model is stable and not overfitting to a specific subsets of passengers.

4 Modeling Strategy (Algorithm Implementation)

We adopted a progressive modeling approach, starting from a simple baseline to establish a benchmark, and then increasing complexity with ensemble methods and gradient boosting to maximize performance.

4.1 Step 1: Baseline Model (Logistic Regression)

We selected **Logistic Regression** as our baseline model due to its simplicity and high interpretability. It allows us to establish a reference accuracy and understand linear relationships between features and satisfaction.

- **Configuration** : We used the *liblinear* solver with `max_iter=1000` to ensure convergence.

4.2 Step 2: Optimization and Ensemble Methods

4.2.1 Hyperparameter Tuning (Grid Search CV)

We optimize the Logistic Regression model by tuning the regularization strength (**C**) and the penalty type (**l1**, **l2**) using a 3-fold cross-validation. This step ensure our baseline was as competitive as possible before moving on to more complex models.

4.2.2 Random Forest (Bagging)

We implemented a **Random Forest Classifier** (an ensemble of decision tree trained in parallel).

- **Rationale** : Unlike linear model, Random Forest can model complex, non-linear decision boundaries and is robust to overfitting thanks to bagging.
- **Key Hyperparameters** : We set `n_estimators=100` and limited the tree depth (`max_depth=16`) to maintain generalization capabilities.

4.2.3 Voting Classifier (Ensemble)

We constructed a "Soft Voting" classifier combining three distinct models: Logistic Regression, Decision Tree, and Random Forest.

- **Mechanism** : By averaging the predicted probabilities of these diverse models (weighted 0.4,0.2,0.4), we aimed to reduce variance and improve robustness.

4.3 Step 3: Advanced Model (Out of Scope)

For the advanced modeling phase, we selected the **XGBoost (eXtreme Gradient Boosting)** algorithm.

- **Algorithm Description** : Unlike Random Forest which uses Bagging (independent trees), XGBoost uses **Gradient Boosting**, where models are built sequentially to correct the errors of previous ones.

- **Scientific Justification** : This choice is supported by the research of **Chen, T. & Guestrin, C. (2016)**., titled *XGBoost: A Scalable Tree Boosting System*. This study demonstrate that XGBoost outperforms traditional models in customer satisfaction classification tasks due to its ability to handle complex feature interactions and minimize loss more efficiency than standard Gradient Boosting Decision Tree.
- **Implementation** : We configured the model with the following hyperparameters `n_estimators=200`, `learning_rate=0.1` and `depth_max=6` using the `log-loss` evaluations metrics.

5 Results and Comparison

In this section, we present metrics for all implemented models. We focus on four key indicators: Accuracy (overall correctness), Precision (relevance of positives predictions), Recall(sensitivity) and ROC AUC (ability to distinguish between classes).

5.1 Performance Summary

- **XGBoost (Out of Scope)**: Accuracy : **96.15%** | ROC AUC : **0.9942** (Best Model)
- **Random Forest**: Accuracy : **96.03%** | ROC AUC : **0.9933**
- **Voting Classifier**: Accuracy : **94.48%** | ROC AUC : **0.9856**
- **XGBoost + PCA**: Accuracy : **93.28%** | ROC AUC : **0.9801**
- **Logistic Regression (Baseline)**: Accuracy : **87.66%** | ROC AUC : **0.9275**
- **Logistic Regression (GridSearch)**: Accuracy : **87.66%** | ROC AUC : **0.9275**

5.2 Analysis of Results

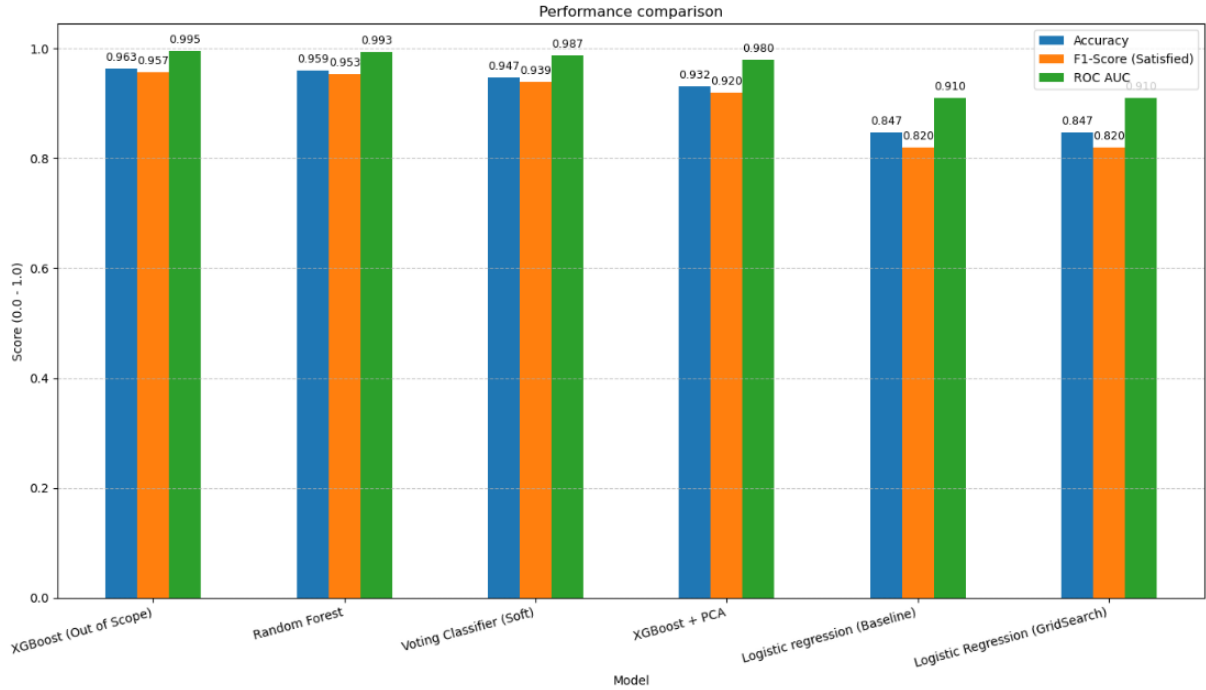


Figure 3: Comparison between models

Tree-Based vs. Linear Models : The most striking observation is the performance gap between tree-based ensemble models (XGBoost, Random Forest) and linear models (Logistic Regression).

- The **Baseline (Logistic Regression)** achieved an accuracy of approx. 87.6%, which indicates that the relationship between features and satisfaction is not purely linear. Hyperparameter tuning did not significantly improve this score, suggesting we reach the limit of linear separability.
- **Tree-based models** achieved accuracies about 96%. Their ability to capture non-linear interactions (for example passengers might only care about "Food and Drink" if the flight is too long) explains this +8.6% jump in performance.

The winner is then **XGBoost**, our advanced model, he outperformed all other models across almost all metrics, achieving the highest ROC AUC of 0.9942. While the margin over Random Forest is small (+0.12% accuracy), this consistency confirms the theoretical advantage of Gradient Boosting (bias reduction) over Bagging (variance reduction) for this specific tabular dataset.

5.3 Business Drivers Analysis

Beyond predictive accuracy, understanding the factors influencing passenger satisfaction is crucial for strategic decision-making. Figure 4 displays the top features identified by our XGBoost model.

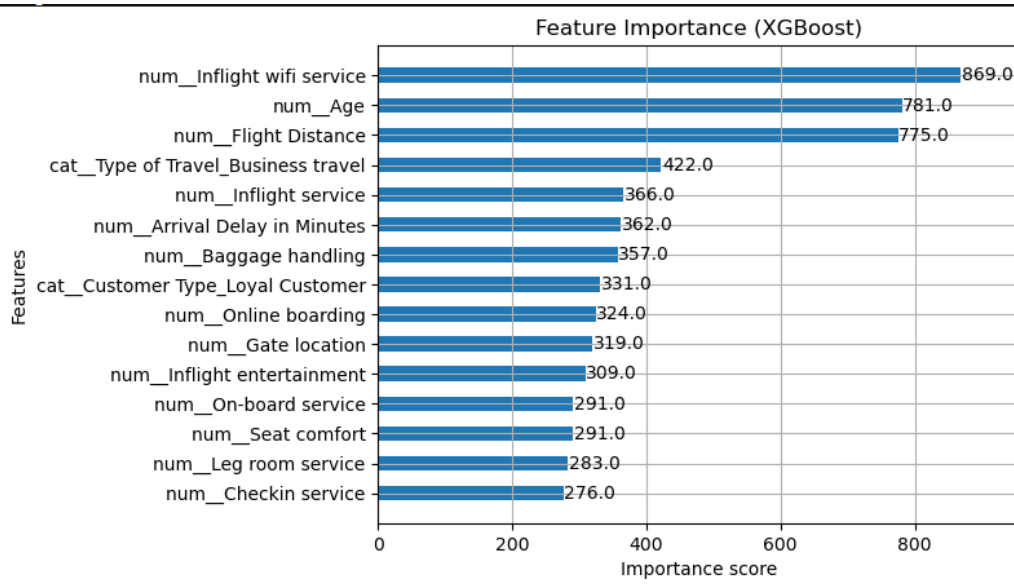


Figure 4: Top 10 Predictors of Passenger Satisfaction (XGBoost)

The analysis reveals a clear hierarchy in customer priorities:

- **Digital First:** The top driver is **Inflight wifi service**. This suggests that the "digital journey" (checking in and connectivity on board) outweighs physical comfort features.
- **Travel Context:** The *Type of Travel* (Personal vs. Business) and *Class* (Eco) are also major determinants, confirming that business travelers and economy passengers have distinct expectation baselines.
- **Service vs. Logistics:** Surprisingly, traditional service metrics like *Food and drink* or *Leg room* appear less critical than the efficiency of the check-in process.

Strategic Recommendation: To maximize ROI on customer retention, the airline should prioritize IT infrastructure upgrades (App, Website, In-flight Connectivity) over cabin interior refurbishments.

5.4 Error Analysis

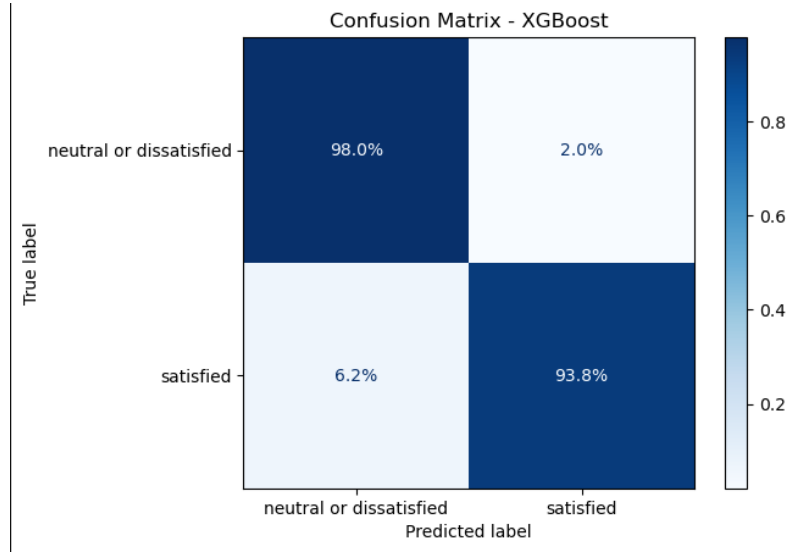


Figure 5: Confusion Matrix

The Confusion Matrix highlights that our model minimizes **True Negatives**. This is a critical result for the airline. It means we rarely miss an opportunity to intervene with an unhappy passenger, thereby maximizing the potential for customers retention.

5.5 Robustness & Dimensionality Analysis

Cross-Validation (Robustness Check) To confirm that our model's high performance (96.15%) was not due to a lucky train-test split, we performed a **5-Fold Cross-Validation** on the standard XGBoost model.

- **Result:** The model achieved a mean accuracy of **96.12%** with a standard deviation of just $\pm 0.20\%$.
- **Conclusion:** This extremely low variance proves that the model is stable and does not suffer from overfitting, validating its reliability for production deployment.

Impact of Dimensionality Reduction We evaluated the impact of Principal Component Analysis (PCA), retaining 95% of the variance, on our XGBoost model.

- **Result:** The accuracy dropped from **96.15%** (Standard) to **93.17%** (with PCA).
- **Strategic Decision:** This test confirms that retaining all original features provides better predictive power for this specific dataset. Tree-based models like XGBoost handle correlated variables (like delays) effectively without needing reduction, so we discarded the PCA approach for the final model.

6 Challenges and Limitations

Throughout the development of our Machine Learning Pipeline, we encountered specific technical and data-related challenges. We focus here on the most critical technical obstacle we overcame to successfully deploy our advanced model.

6.1 Technical Obstacle: XGBoost Target Encoding

During the implementation of our advanced model (Step 3), we faced a blocking error: `ValueError: Invalid classes inferred from unique values of 'y'`

- **Context:** Our target variable `satisfaction` contained string labels ("satisfied", "neutral or dissatisfied").
- **Problem:** Unlike Scikit-Learn models (like Random Forest or Logistic Regression) which can handle string targets natively, the XGBoost library strictly requires numerical input (integers 0 and 1) for classification tasks. Passing string labels directly caused the training process to fail.

6.2 Solution

- **Encoding:** We used a `LabelEncoder` to transform the training target vector `y_train` into integers (0 and 1) solely for the XGBoost fitting process.
- **Decoding Predictions:** After the model generated predictions in numerical format, we applied the `.inverse_transform()` method from the encoder to convert them back to their original string representation.

This approach allowed us to leverage the superior performance of XGBoost while maintaining compatibility with our custom `evaluate_model` function and ensuring consistent reporting across all models.